12 Which statement about geostationary orbits on Earth is **false**?

A A geostationary orbit must be directly above the equator.

B All satellites in geostationary orbit must have the same mass.

C The period of a geostationary orbit must be 24 hours.

D There is only one possible radius for a geostationary orbit.

Ans: **B**

Option A: Since orbits can only be centred about the centre of mass of the Earth, only orbits around the equator will allow for geostationary orbit.

Option B: untrue. Mass of satellite doesn't affect the period.

$$\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$$

$$T^2 = \frac{4\pi^2}{GM} r^3 \text{ (independent of mass of satellite)}$$

Option C: Since the period of the Earth's rotation is 24 hours, that must also be the period of the orbit.

Option D: Looking at the math in option B answers, $T^2 \propto r^3$. This means that every period corresponds to a specific orbiting radius.